

AI-Assisted Contract Intelligence for Enterprise Lifecycle Management: An Integrated Analysis of Contract Risk, Semantic Retrieval, and Approval Workflows

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Abstract— Large enterprises rarely deal with only a few contracts: more often than not, they are handling hundreds or thousands simultaneously and managing customer, vendor, supplier, and other business partners. In the traditional approach these documents are shuffled through departments, kept in various, disconnected repositories, and manually examined, making it difficult to ascertain the current status of any individual contract at any given time. The Enterprise Contract Lifecycle and Approval Management System introduced here aims to consolidate that process on a single platform, ranging from the initial upload and analysis to approval, execution, renewal, and eventually an archiving. It combines document processing, AI and natural-language methods, duplicate detection, clause identification, risk scoring, approval roles, retrieval-augmented generation, version tracking, and audit logging into a single operational system. Once uploaded, the contracts are processed to extract information such as type of agreement, parties to the agreements, relevant dates, amounts of money, obligations and specific clauses. A classification model will predict the nature of the relationship the contract embodies. Moreover, there will be many algorithms that can detect duplicates, both in terms of identical contracts and contracts that have been reworded to disguise their meaning. Clauses are next reviewed and risk is assessed using a combination of rule-based checks, NLP, and AI support. Based on such factors as the type of relationship, risk class, and contract value, the system can be set up to design an approval workflow and route

the contract through the relevant approvers. A retrieval-augmented generation module enables asking questions about contracts that are permitted to view, without crossing a role-based access boundary. A document management system can track all versions of the new upload, changes, and history of the document, while a cryptographic audit trail can detect tampering after the fact. A centralized dashboard allows you to monitor contracts, approvals, obligations, and renewals throughout the life cycle

Index Terms— Contract Lifecycle Management, Artificial Intelligence, Natural Language Processing, Contract Classification, Duplicate Detection, Risk Analysis, Retrieval-Augmented Generation, Approval Workflow, Role-Based Access Control, Audit Trail.

I. INTRODUCTION

Contracts are the backbone of the organization today: they formally create the legal relationships between a company and its customers, vendors, suppliers, employees, and partners. One enterprise is expected to handle hundreds or thousands of such agreements simultaneously, each with their obligations, financial figures, window for renewal, and approval requirements. It directly impacts the operational efficiency, compliance, risk, and the quality of decisions that can be made based on them.

Contract Lifecycle Management (CLM) is the process of managing contracts from the drafting and review through

approval, execution, monitoring, renewal, archiving. Research in legal NLP has shown that AI can assist in a number of these tasks: classifying clauses, extracting evidence, reasoning across the entire document, and more. Public datasets such as CUAD and ContractNLI have been instrumental in the development of automated contract analysis.

That is, however, only half of the problem. A practical enterprise solution should couple what is extracted from contract text to the workflows that the contract will go through. Depending on the type, risk, value, or the internal policy, a contract may need to go through a set of departments before being finalized. Therefore, a CLM platform worth building should not only read the contract but act on what it reads to drive the approval workflow and subsequent stages.

The system designed to address this problem brings a number of the components discussed earlier together under the umbrella of a single solution. It starts right on the upload of a contract: document-processing routines extract the text and, in addition, retain its structure, and an AI component determines what kind of relationship the contract represents, assigning a confidence and the supporting evidence.

There exist several levels of duplicate detection within this system; namely, exact duplicates are identified through cryptographic hashing, near duplicates sharing some metadata or wording are identified as derivative contracts, and finally, semantic embeddings discover semantically different, yet similar contracts to identify differing phrasings of the same contract.

Having analyzed the document, the program extracts the relevant clauses and classifies them into categories. The clause taxonomy uses CUAD's structure as the reference to define what constitutes an important contractual provision: CUAD is a dataset built of 13,101 labelled clauses across 510 commercial contracts, covering 41 categories, pointing to the value a well-structured dataset can have in training automatic review tools.

Risk scoring pulls together the rules, clauses extracted from the document, contract values, key dates, stated obligations and AI guided interpretation. Rather than simply returning a risk score, the system ties every flagged risk to the clause or a piece of document evidence that leads to it.

At that point, the contract can be submitted for approval: the workflow proceeds sequentially through predefined roles: Legal, Manager, Finance, and Compliance, for example, and only the step that is currently in progress is available for action, while subsequent steps can be seen but locked to the user.

A retrieval-augmented generation (RAG) module further allows asking questions about the contracts that are permitted

to view. The documents are broken down into semantically coherent chunks that are converted to vector embeddings and kept for similarity search, and for any question, the pertinent chunks are retrieved and passed to the language model to ensure that its answer is grounded on the contract text.

To complete the lifecycle, version control, monitoring of renewals and obligations, a real-time dashboard, role-based access constraints and an audit log to spot tampering are all part of the solution. The idea is to manage contract intelligence plus workflow as an integrated system and not two separate tools which are bolted together.

II. LITERATURE REVIEW

Since a great deal of legal information is textual, AI and NLP have taken a prominent role in legal technology. Researchers in the field of legal NLP have explored document classification, clause extraction, semantic similarity, question answering, and natural language inference.

Hendrycks et al. created the Contract Understanding Atticus Dataset (CUAD) with the purpose of automating contract review [1]. It provides expert annotations of 41 clause categories drawn from commercial contracts and has become a standard benchmark for clause extraction as well as contract-related question answering.

Koreeda and Manning have assembled ContractNLI, which focuses on document-level natural language inference within contracts [2]. It matches contracts with hypotheses that a system must classify as supported, contradicted, or unmentioned, supported by evidence ranges.

Chalkidis et al. have developed LEGAL-BERT, a language model fine-tuned on legal text. [3] Their results highlight the value of domain-specific pre-training and the importance of the lexicon and phrasing used.

The same team later introduced LexGLUE, a benchmark that covers a number of legal datasets and tasks [4]. It demonstrates the difficulty of one language model to generalize over a variety of legal problems.

Reimers and Gurevych have presented Sentence-BERT, which generates sentence embeddings optimized for semantic similarity and retrieval [5] — relevant here, since two contracts may contain the same substance in different wording and must be identified as similar.

Devlin et al.'s BERT demonstrated the power of bidirectional Transformer representations for language understanding [6], and has become the base for a number of specialized models, including the legal ones.

Vaswani et al.'s "Attention Is All You Need" paper introduces the Transformer architecture that underlies all of the subsequent work [7].

Lewis et al. have introduced retrieval-augmented generation, which combines a retrieval step with text generation to allow a model to incorporate external information before answering [8] — a natural fit for question answering based on a particular document.

Provision classification has also seen attention in legal NLP: datasets such as LEDGAR include large volumes of contractual provisions and demonstrate that supervised learning can categorize legal text [9], while LexGLUE incorporates classification tasks to its suite [10].

More recently, large language models have been applied directly to clause extraction, summarization, risk flagging, question answering, and contract reasoning [11, 12]. However, survey work in this field always draws attention to the same stumbling blocks: long documents, dense domain vocabulary, interdependent clauses, and the need for decisions that come with supporting evidence (rather than simply a verdict). Overall, this research provides a solid foundation for individual components of the solution puzzle — legal language understanding, classification, semantic similarity, clause extraction, question answering. What it does not address is how these components can be integrated with authentication, role-based permissions, approval routing, version history, renewal tracking, obligations, audit logging, and live lifecycle visibility; all of which a real enterprise deployment must involve.

The system proposed here builds upon this foundation, but does not claim to reinvent any of the individual capabilities. The contribution is in the integration, linking AI-based contract comprehension, risk evaluation, sequential approval, RAG-based querying, version management, and auditable lifecycle management.

III. METHODOLOGY

The system design is grouped into a life cycle that includes document processing, AI-enabled contract understanding, duplicate identification, semantic search, clause discovery,

risk evaluation, process control, and lifecycle auditing. Fig. 1 below represents the overall process flow of the project beginning with its initiation and ending with its closure. Fig. 2, which is provided further below, illustrates the overall architecture of the project – the client layer, the application/API layer, core business services, the AI and document-intelligence layer, and the data/infrastructure layer at the bottom.

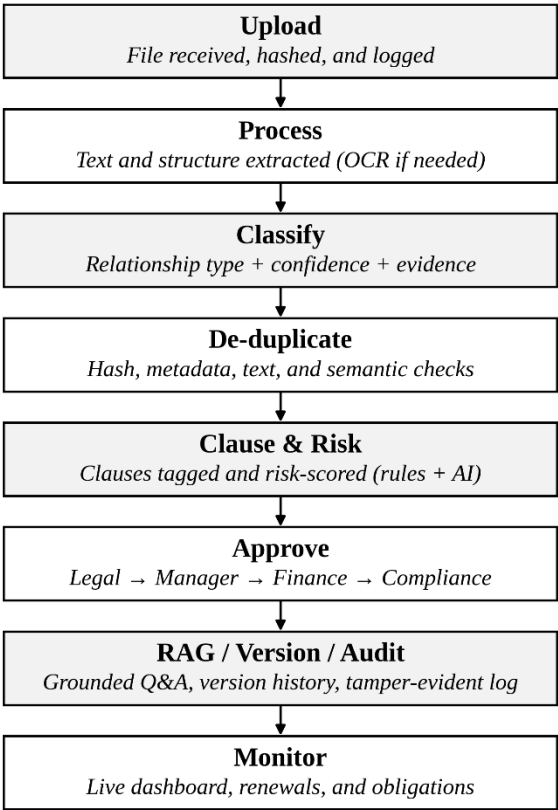
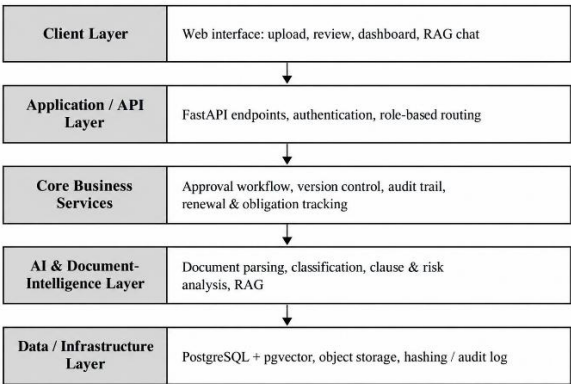


Fig. 2. Overall Workflow of the Enterprise Contract Lifecycle and Approval Management System.

Fig. 1. architecture of the contract lifecycle system.



A. Contract Upload and Document Processing

The contract upload starts the workflow, with the file being uploaded by an authorized user via web interface. Document formats supported: PDFs and other text-based formats. The file is safely stored in the backend. The file’s relevant fields are also stored in the database immediately upon upload: file name, size, upload timestamp, user who uploaded, and a cryptographic hash. For PDFs: text extraction and processing utilizes document-processing libraries, and Azure Document Intelligence when needed. OCR is applied to scanned pages, while page layout is preserved to maintain the page’s context. The following gets extracted for the contract: title, contract number, parties, counterparty, effective date, expiry date, renewal details, contract value, currency, payment terms,

obligations, important clauses, document pages, and document sections. This step is performed asynchronously and hence is invisible to the UI and saves the front end from timeout issues caused by huge documents.

B. Contract Relationship Classification

The contract relationship classification is an important step in deciding the type of entity involved in the contract. With the textual information extracted, the system analyses the text and entities to identify the best-fitting classification, such as: Vendor, Customer, Supplier, Business Partner, Independent Contractor, Licensing, and Commercial Lease. This step involves performing a semantic analysis of the text using AI/LLM and combining it with the extracted metadata and evidence to classify the contract, returning the contract type, confidence, reason, and evidence. For example, a document that has clauses that involve purchasing a good from an external organization would be classified as a Supplier Agreement. In the case the confidence score is below a threshold, the document would be flagged for human review rather than being positively predicted and then transparently classified with a reason and evidence.

C. Duplicate Detection

Duplicate detection can be performed on different levels of comparison: Exact Duplicate Detection: A hash (SHA-256) is assigned to each uploaded file. Files that have the same hash are considered to be exact copies of each other. Other Metadata-Based Detection: Contract numbers, titles, parties, dates effective/expiry and values can be used to detect duplicates Text Similarity: The text of the contracts is normalized and compared using TF-IDF representations and cosine similarity. Semantic Similarity: Embeddings of contract chunks/documents allow for similarity searches in vector space and finding semantically similar contracts with different wording. By taking all four levels into consideration for the purpose of detecting duplicates, the issue of files having the same hash value is bypassed.

D. Semantic Chunking and Embedding Generation

Instead of treating a large document as one continuous stream of text, it is chunked into semantically coherent segments, with consideration to headings, section breaks, clauses, paragraph boundaries and content flow. Each chunk is tagged with metadata such as: contract ID, version ID, page number, section, and chunk index, before being embedded and stored in PostgreSQL using the pgvector extension. Sentence-BERT's dense sentence representations [5] research demonstrates why this approach is effective for similarity search/retrieval, and this vector store powers the RAG module described later.

E. Clause Classification

Important clauses are extracted from the processed contract using a taxonomy inspired by CUAD's 41 clause categories

[1]. The categories that can be recognized by the system include the following: Agreement Date, Effective Date, Expiration Date, Renewal, Termination, Termination for Convenience, Confidentiality, Non-Compete, Non-Solicitation, Intellectual Property, Assignment, Governing Law, Liability, Indemnification, Insurance, Warranty, Payment Terms, Audit Rights, Data Protection, and Dispute Resolution. Each clause that the system detects can be stored together with its type, extracted text, page number, evidence, and relevance to overall risk, with traceability back to the actual text in the document from the AI's result.

F. Risk Analysis

Risk is assessed using a combination of business rules and AI/NLP semantic analysis. The business rules concern such issues as expiry date omission, short termination notice, high value, unlimited liability, missing of required insurance, confidentiality clause absence, inappropriate payment terms, automatic renewal and high-risk termination conditions. The AI/NLP parses the context of each clause in order to evaluate its impact. Combining these signals identifies an overall risk category (Low, Medium, High, or Critical) and the individual findings are stored along with their risk category, severity, description, clause, page number, evidence, and recommendation.

G. Sequential Approval Management

The contract analysis can be sent for approval through the sequential approval workflow (usually Legal → Manager → Finance → Compliance, as shown in Fig. 3), where initially only the Legal status is ACTIVE, while the rest are WAITING. After Legal's approval, the Legal status changes to APPROVED, Manager becomes ACTIVE, and the rest are WAITING. Then, after the Manager's approval, Finance becomes ACTIVE, and the others are WAITING. Next, after Finance's approval, Compliance becomes ACTIVE, and the contract is finally approved. Only the step that is active can be performed, and the rest are greyed out until the previous step is done. It is done at the back-end, and it cannot be achieved by simply disabling the button on the front-end. The approver can approve, request changes, or reject the contract. If changes are requested, the contract can be sent back for changes, and resubmitted for approval. Therefore, a new version of the contract can be uploaded to the same contract record instead of creating a new contract. Fig. 3. Sequential approval workflow: only the approver can perform an action, and the rest are greyed out until the previous step is done.

H. Role-Based Access Control

The Role-Based Access Control (RBAC) will guarantee that users may only access the functions and data that relate to their positions within the company. The users will be divided into groups depending on their roles: Contract Administrator, Legal, Manager, Finance, Compliance, Department User, and System Administrator. Notably, the front end and the back

end will be modified according to the roles. The FastAPI backend will be used to authorize and authenticate all the requests before any sensitive data is released to the users. Database-level access controls are an additional layer of control.

I. Retrieval-Augmented Generation

A retrieval-augmented generation module provides users with the ability to ask questions about the contract content. A question is converted to an embedding, and the vector database is queried for relevant chunks, which are then provided to the language model as context: User Question → Query Embedding → Vector Search → Relevant Contract Chunks → LLM → Context-based Answer. Answers can reference the specific contract section/page from which they originate. Role-based filtering occurs before retrieval commences, so users are never able to access content from a contract they should not see. RAG, as first proposed [8], is a combination of a parametric language model and external knowledge retrieval, ideal for question-answering scenarios anchored to specific documents.

J. Version Control

Contracts undergo multiple changes during negotiations, reviews, and renewals, so instead of overwriting a file, a new version is created (e.g., v1.0 to v1.1 to v2.0, etc.). The version number, upload timestamp, user who uploaded, file hash, description of changes, reference to the previous version, and whether the version is current are stored. A comparison view highlights the differences between two versions, and risk levels can be compared across versions to ascertain if a contract is becoming more or less risky over time.

K. Renewal and Obligation Tracking

Many contracts have ongoing obligations or renewal clauses which require our attention after their approval. The system stores the information on relevant dates (effective, expiry, renewal, notice period) and can remind us when expiry or renewal date is coming up. Each obligation is stored together with a description, person responsible, deadline, status, and priority level

L. Audit Trail

All actions performed with the contract that require approval are stored including: upload, analysis, classification, submission, approval, rejection, request for change, new version, renewal, and archival. The audit system uses SHA-256 hashing to create a chain, proving any tampering, linking to the record it references and describing the action, similarly to the following: $\text{Hash}_n = \text{SHA-256}(\text{Action}_n + \text{PreviousHash})$ (1) Where each next hash is dependent on the previous one, thus changing any data from a record will invalidate all hashes from that record onwards, proving tampering. M. Monitoring

A central monitoring dashboard provides an overview of the number of contracts, drafts, contracts in processing, items ready for review, pending approvals, approved and active contracts, rejections, high-risk contracts, approaching expirations, renewals due, and open obligations. Approval counts can also be broken down by role. Because the dashboards update in real time, there is no need to refresh the page to see the latest status.

IV. RESULT AND ANALYSIS

When viewed as a whole the system provides an integrated approach to managing contracts across their lifecycle, combining document processing, AI, NLP, similarity analysis, workflow automation, and secure lifecycle handling. The upload module is where documents are stored and processed, with the processing pipeline extracting both text and metadata as soon as a file arrives. Classification spares the user the effort of tagging a document as being a vendor, supplier, customer, or partner relationship, as the system reads the contract and returns a predicted relationship along with confidence and evidence. Duplicate detection works on multiple levels at the same time, with SHA-256 hashing identifying exact file matches, with metadata and text similarity extending the coverage, and semantic embeddings detecting contracts with the same meaning but different wording. The clause analysis performs a transformation of text into the information of relevance to specific provisions that address particular issues, akin to how CUAD demonstrated the process could be automated through categories of clauses supported by evidence [1]. The risk analysis is based on deterministic rules as well as AI/NLP signals, and the results are linked to relevant clauses/pages, not just a score, making it more actionable than a generic score. The approval module processes each contract through its own lifecycle, where processing of the next step is only triggered by the completion of a preceding step, and processing of later steps is never performed before earlier ones. The RAG component identifies relevant chunks of information in the vector DB before returning the response, thus ensuring that the response is based on the contract text rather than the LM's general knowledge. The version control module prevents any overwriting of information by keeping an archive of every previous version of a contract with timestamps, letting you see what changes were made and compare between the different versions of a contract. The audit mechanism uses the invariant cryptographic hashing of the entire set of records to create a tamper-evident record of events; any tampering will be revealed by later changes in the chain of hashes. The dashboard is a single view of the contract's status, approvals, risk, renewals, and obligations; because it is dynamic and real-time, it does not require refreshing to display the latest information. Taken together the system demonstrates how AI-driven document understanding and operational workflow management can form a connected process rather than two separate systems bolted together to cover the full contract lifecycle. Functional Evaluation The system can be evaluated with test cases such as those summarized in Table I. TABLE I

Functional Test Cases

Test Case	Expected Result
Upload a new contract	A single contract record gets created
Upload the same file again	The duplicate gets flagged
Upload a contract that is semantically similar	A similarity warning is triggered
Run document processing	Metadata is extracted correctly
Classify a contract	Type, confidence, and evidence are generated
Detect a clause	Clause category and evidence are displayed
Analyze risk	Risk level and findings are generated
Submit for approval	The first configured step becomes active
Legal approves	The manager step becomes active
Manager approves	The finance step becomes active
Finance approves	The compliance step becomes active
Compliance approves	The workflow is marked as complete
Attempt a locked step	The backend rejects the request
Request changes	The contract returns to revision state
Upload a new version	The version number is incremented
Ask a contract question	Relevant context is retrieved
Unauthorized query	Access gets denied
Tamper with an audit record	Hash-chain inconsistency becomes detectable
Contract nears expiry	A renewal/expiry notice is generated

Evaluation should be based on figures actually measured in the production system: accuracy, precision, recall, F1, processing time, duplicate detection thresholds, and user response time are only meaningful if they have been actually measured. Proposed Evaluation Protocol A quantitative

evaluation of the system has been proposed but not conducted yet, and is planned as the next step in this work. It would be based on a test set of 40–60 contracts in the seven relationship categories described in Section III-B, obtained from publicly available, de-identified sources, with additional 8–10 near-duplicate pairs inserted as test cases for the duplicate detection pipeline. Gold-standard labels for relationship types and clause categories would be obtained from two or three annotated using the taxonomy in Section III-E, and agreement between annotators would be measured with Cohen’s or Fleiss’ kappa. Using these labels, the evaluation protocol described in this section would report: (i) accuracy and macro-F1 of relationship-type classification; (ii) precision, recall, and F1 of the detection module at each of the comparison levels described in Section III-C; (iii) precision, recall, and F1 of clause categorization for each clause category; (iv) agreement between the system’s risk category and that of a human reviewer; (v) end-to-end contract processing time; and (vi) answer groundedness for the retrieval-augmented generation module, defined as the proportion of answers for which a human reviewer confirmed that the answer’s supporting context chunk actually supports the answer. To determine the contribution of the AI pipeline’s innovations, the evaluation protocol would compare the described system with a baseline of keyword-based methods with no learning components, and, if available, a commercial or open-source CLM tool. Actually, performing this evaluation and reporting the figures would be the goal of the controlled experiment following this study. Limitations The evaluation in Section IV is qualitative: it documents that each module performs its intended function under the test cases in Table I, but does not report quantitative measures such as classification accuracy, precision/recall of duplicate detection, agreement with human reviewers in risk analysis, groundedness of answers, or system latency. These metrics require a test set with gold-standard labels, and therefore will be the subject of further research. Additionally, the system has not been benchmarked against commercial or open-source competitors, and therefore makes no claims of its own superiority over existing tools. The clause and risk taxonomies used are based on the CUAD categories but were not validated against their gold-standard labels, and the confidence thresholds for human review routing and similarity-based duplicate detection have not been validated either. Addressing these points would be the goal of the next stage, a quantitative evaluation as described above.

V. CONCLUSION

The Enterprise Contract Lifecycle and Approval Management System, proposed in this paper, offers a unified solution for enterprise contract lifecycle and approval management. This paper presents a system that integrates document processing, artificial intelligence, natural language processing, duplicate

detection, clause classification, risk analysis, sequential approval, role-based access control, retrieval-augmented generation, version control, renewal monitoring, obligation tracking, and audit management within the same platform. Employing multiple duplicate-detection techniques in combination increases the likelihood of detecting both exact and near-duplicates, and AI-assisted classification provides a relationship label with confidence scores and basis, rather than a simple guess. The sequential approval engine is the link between the contract analysis and the actual decision in the organization — the contracts move through the relevant approval steps and the backend enforces that none are taken out of order. The RAG component enables users to obtain grounded, context-based answers from contracts they are permitted to view. The version control keeps a complete record of document changes, and the renewal and obligation tracking addresses what still needs to be addressed after a contract has been approved. In general, the program implies the combined management of enterprise contracts, not separating contracts analysis, risk assessment, approval, and monitoring but bringing them together in one program, thus increasing the visibility, efficiency of operations, control, and decision-making quality in general.

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